# HW4 – Clustering Analysis

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February 12, 2026

## 1 Introduction

The objective of this assignment is to apply and compare multiple clustering algorithms on multidimensional data. The multidimensional data is derived from the Spotify/Youtube dataset. The selected features are Liveness, Energy, and Loudness.

In Part 2 of this assignment, the three-dimensional density visualization is constructed using the Iris dataset. This visualization showcases the frequency of repeated feature combinations and provides an alternative way to explore structure in the data.

## 2 Data Preprocessing

For the Spotify dataset, only the Liveness, Energy, and Loudness columns were utilized. Rows containing missing values were removed to ensure valid distance calculations.

For the Iris dataset, petal length and petal width were extracted. Duplicate coordinate pairs were counted to construct a third dimension representing density. This axis that is part of the three dimensional structure of the visualization, is the Z axis, where this count is existent.

## 3 K-Means Clustering

The K-Means elbow method was used to estimate an appropriate number of clusters. Figure 1 shows inertia as a function of the number of clusters. A sharp decrease in inertia occurs for small values of K, followed by a gradual leveling off around K = 6 to K = 8. This suggests diminishing returns beyond this range, where the data naturally partitions in 6 to 8 clusters.

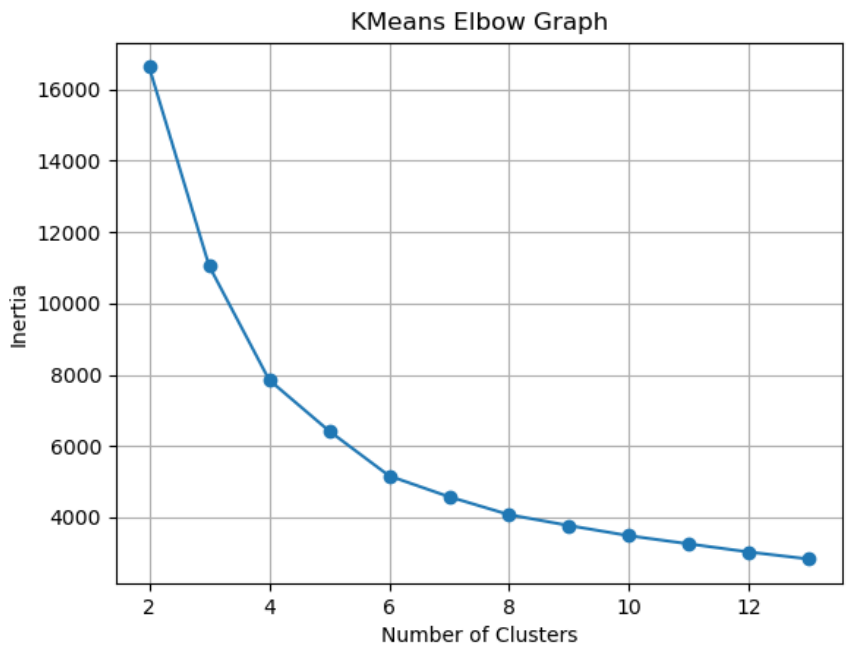


Figure 1: K-Means Elbow Graph

# 4 KMeans Scatter Plot

In Figure 2, this represents the scatter plot with the optimal number of k=6, demonstrated by the KMeans elbow graph. KMeans partitions the Spotify dataset into six clusters based on similarity in Liveness, Energy, and Loudness. The clusters appear along the Energy and Loudness axes. Because KMeans minimizes cluster variance, the groupings are compact.

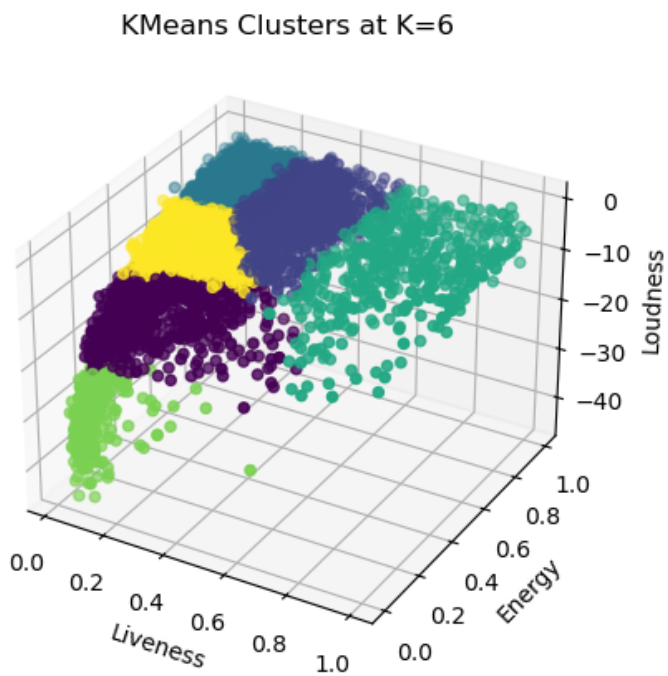


Figure 2: KMeans Scatter

# 5 DBSCAN Clustering

DBSCAN primarily concerns clustering with density and is an algorithm that relies on two parameters; the radius or $\varepsilon$ and the minimum number of points required to form a dense region. Figure 3 presents the elbow graph for minimum k = 5 points.

The plot shows long range of dense points, followed by a sharp increase in distances. The elbow occurs just before this steep increase and as a result, showcases an appropriate radius or $\varepsilon$. Points beyond this threshold are most likely outliers, while points within the long range are core points. This suggests that the Spotify/Youtube dataset may contain data that is fairly dense, with few outliers.

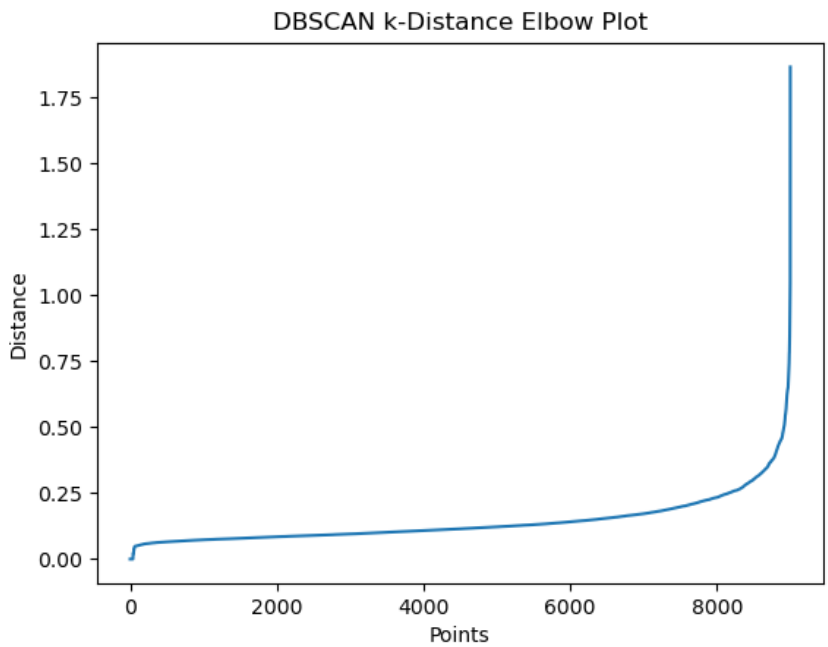


Figure 3: DBSCAN Elbow Graph

# 6 DBSCAN Clustering Scatter Plot

Using min_points = 7, the algorithm detects one dominant dense region. Points that are shaded are outliers within the dataset. This demonstrates DBSCAN's ability to detect outliers.

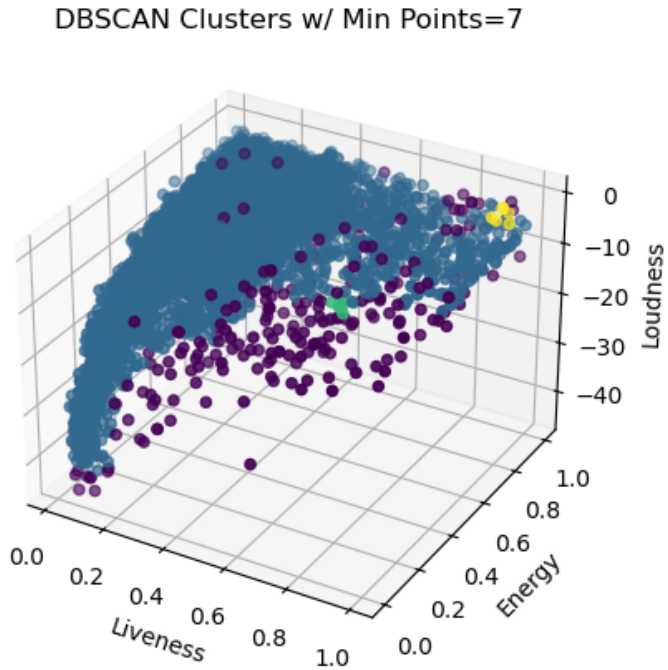


Figure 4: DBSCAN Scatter

# 7 Mean-Shift Clustering

Mean-Shift clustering was applied using a range of bandwidth values. Figure 5 shows the number of detected clusters as a function of bandwidth. At small bandwidth values, a large number of clusters is detected, indicating sensitivity to local variations in the data.

As the bandwidth increases, clusters merge and the number of clusters decreases. A more stable region is between bandwidth values of approximately 0.8 and 1.2. This range represents a scale where dense regions in the data form stable groupings. Mean-Shift in this case identifies clusters as modes of the estimated density function.

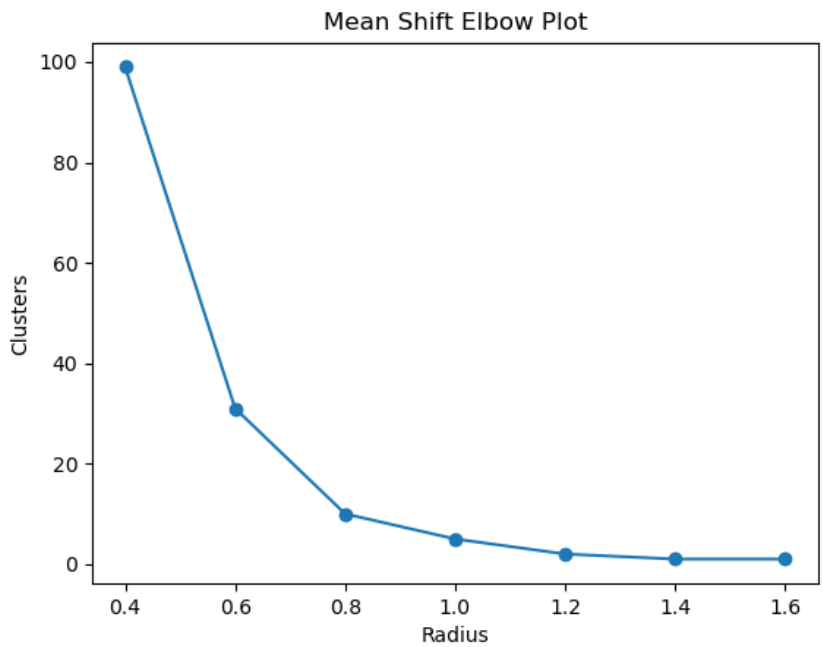


Figure 5: Mean-Shift Elbow Graph

# 8 Mean-Shift Clustering Scatter Plot

Mean-Shift clusters the data by shifting points toward density peaks. Compared to KMeans, Mean-Shift generates smooth density, and determines the number of clusters based on density structure.

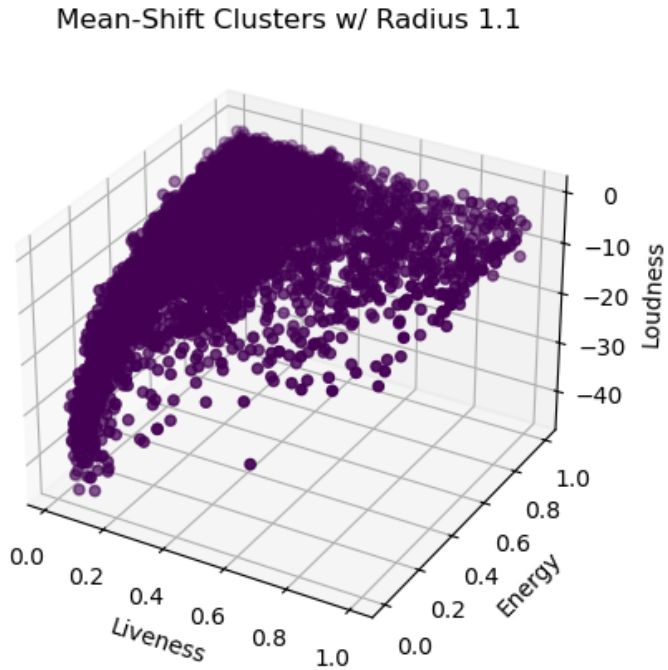


Figure 6: Mean-Shift Scatter

# 9 Summary of Clustering Methods

K-Means provides a simple baseline and considers defined number of clusters in order to compute. DBSCAN and Mean-Shift are both density framed methods that adapt more naturally to the underlying structure of the data.

# 10 Iris Dataset 3D Density Visualization

Figure 7 shows a three-dimensional density map of the Iris dataset. Petal length and petal width form the horizontal axes, while the vertical axis represents the frequency of duplicate features. The visualization showcases regions of high density that correspond to more common sets of (x,y) values.

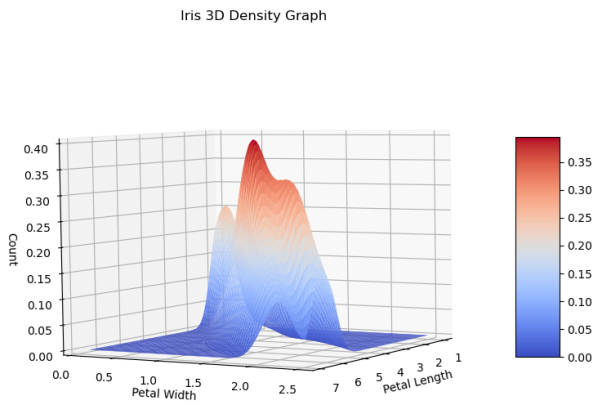


Figure 7: 3D Density Graph for the Iris Dataset